

Pig in a Poke^{*}

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Abstract

We study how market competition interacts with the design of buyers' information about product quality when sellers privately choose quality. We characterize seller-optimal information under monopoly and in symmetric equilibria of Bertrand competition. Under competition, optimal disclosure is partially informative and sustains randomization over quality. It differentiates the buyer's beliefs about competing products, softening price competition. Under monopoly, full disclosure is seller-optimal: it sustains efficient quality provision and allows the seller to extract the resulting surplus. We therefore highlight that, when disclosure is designed in sellers' interests, competition can lower quality and total surplus.

Keywords: Quality, Information Design, Monopoly, Bertrand Competition.

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1. INTRODUCTION

According to a folk story, the idiom of buying a “pig in a poke” goes back to medieval marketplaces, where sellers would offer piglets in closed bags (“pokes”) and dishonest sellers would substitute worthless cats. The idiom survives as a warning against buying what one cannot inspect. Why, then, do competing sellers favor a market institution that conceals product quality? In this paper, we show that under symmetric competition, sellers can benefit from opacity. A monopolist profits from opening the bag; competing sellers profit from keeping it closed.

This paper studies how market competition interacts with the design of buyers’ information about product quality when sellers privately choose quality. Contrary to the standard intuition that competition promotes transparency and higher quality, our model shows how it can favor opacity and lower quality. In our model, full transparency is seller-optimal under monopoly, along with any sufficiently informative experiment, and induces efficient high quality. Under competition, seller-optimal information is only partially informative and induces sellers to randomize over quality. When information provision is aligned with sellers’ interests, competition therefore reduces quality and efficiency relative to monopoly.

Formally, we consider a market in which a single buyer purchases from either one seller or two competing sellers. Sellers privately choose the quality of their product prior to trade. High quality is worth one to the buyer and entails an opportunity cost $c \in (0, 1)$ when sold; low quality has zero value and zero cost. The buyer does not observe quality but receives information through an experiment whose signals are publicly observed.¹ Sellers then compete in prices given the buyer’s posterior beliefs. We characterize perfect Bayesian equilibria, where we restrict attention to symmetric equilibria and symmetric, independent experiments in the baseline model. A key feature of the environment is that the prior is endogenous: the buyer’s ex-ante belief that a seller’s quality is high is generated by the sellers’ equilibrium quality choices, which the information structure itself must sustain.

In the absence of competition, transparency is seller-optimal and induces the efficient provision of high quality. A monopolist extracts the entire expected surplus by pricing at the buyer’s expected valuation, and this surplus is largest when a high-quality product is traded with certainty. A sufficiently informative experiment reveals a deviation to low quality and thereby makes

¹The experiment is not chosen by a player in the game, and we characterize which experiment maximizes seller profits. For a discussion on the exogeneity of the information structure see Section 3.

high quality incentive compatible. Full disclosure is one such experiment. The seller's interest in information provision is thus aligned with efficiency.

Competition overturns this logic. If both sellers choose high quality with certainty, Bertrand competition drives prices to marginal cost; if both choose low quality, the buyer is not willing to pay a positive price. Positive profits therefore require sellers to randomize over quality, which in turn requires the information structure to keep them exactly indifferent between the two qualities. Neither full disclosure nor no disclosure can sustain such randomization. A seller-optimal information structure must instead be imperfectly informative to sustain sellers' randomization, and it must differentiate the buyer's beliefs to soften price competition. The indifference requirement distinguishes our problem from information design with an exogenous prior, and it determines the shape of the solution.

Our main result establishes existence and characterizes the seller-optimal information structure. An optimum always exists, is partially informative, induces sellers to randomize over quality, and yields positive profits. When high quality is expensive so that incentives to provide it are weak, the optimal structure takes a simple form: it sometimes reveals a seller's quality perfectly and otherwise provides *sufficient news*: the lowest belief at which a seller can still make a sale. Below this belief, any price acceptable to the buyer is lower than a high-quality seller's cost, so willingness to sell at such a price reveals low quality, and the market breaks down. Perfectly revealing signals help sustain incentives for high quality despite its high cost.

In contrast, when high quality is inexpensive so that incentives to provide it are strong, the optimal information structure retains a continuous range of noisy posterior beliefs. Low quality is never revealed to the buyer, and trade always occurs. This uncertainty keeps low quality profitable even when high quality is cheap.

The optimal experiment transforms quality uncertainty into vertical differentiation in the buyer's posterior beliefs. When the buyer holds different beliefs about the two products, the seller with the higher posterior can charge a premium. Moreover, even when the buyer receives sufficient news about both sellers' products, a low-quality seller earns a positive margin. She cannot undercut as a lower price would reveal her quality. Information asymmetry thus prevents competition from eliminating rents. These profits, however, come at a social cost: relative to the fully informative monopoly benchmark, quality provision is distorted and the allocation is inefficient, as the buyer may purchase a low-quality product over a high-quality one.

From a methodological perspective, we contribute a tractable approach to information design with an endogenous prior. The prior is determined jointly

with the information structure and must be consistent with sellers’ endogenous incentives. Formulating the problem in terms of quantile functions turns the problem into an optimization problem with a linear objective and a convex incentive constraint, so that the optimum is characterized by a Lagrangian argument.

In an extension, we relax the assumption of independent experiments and allow joint experiments that compare the two products directly. A simple “ranking experiment” is then seller-optimal when quality is sufficiently costly: it recommends one seller over the other without perfectly revealing either product’s quality. When qualities differ, the recommendation is more likely to be given to a high-quality product, but sometimes a low-quality product is recommended to keep low quality profitable. The ranking ensures differentiation in the buyer’s beliefs and can therefore enable higher profits than independent experiments. As in the baseline model, sellers benefit from opacity that sustains quality randomization and softens price competition.

Finally, we discuss the robustness and implications of our results. First, the comparison with monopoly relies on equilibrium selection: under full disclosure there also exist asymmetric equilibria in which one seller acts as a monopolist, so that competition need not reduce quality. We argue that the symmetric equilibrium might be the more compelling prediction for ex-ante identical sellers without a coordination device. Second, the analysis speaks to environments where information provision is plausibly shaped by seller interests such as seller-designed marketplaces, certification paid for by sellers, or platforms financed by sellers’ fees. In such environments, competition need not foster transparency. Instead, the interaction of competition and information design can sustain opacity and erode incentives for quality provision, which provides a possible rationale for certification that falls short of fully guaranteeing quality.

The remainder of the paper is structured as follows. Section 2 relates the paper to the literature on information design and endogenous quality provision. Section 3 introduces the model. Section 4.1 analyzes the monopoly benchmark, and Section 4.2 characterizes the seller-optimal information structure under competition. Section 5 discusses correlated information structures, asymmetric equilibria and policy implications, and Section 6 concludes. All proofs are in the Appendix.

2. LITERATURE

We relate our paper to three strands of literature: information design, certification and disclosure of hidden quality, and the interaction between information

and product-market competition.

Information Design Our paper contributes to the literature on Bayesian persuasion and information design (Kamenica and Gentzkow, 2011). We study information design in a market setting between a seller and a buyer. In those applications, the literature distinguishes between the prior over the object of information being either *exogenous* or *endogenous* to the information structure, and between the market being monopolistic or competitive. This distinction is displayed in the following table.

	Exogenous prior	Endogenous prior
Monopoly	Roesler and Szentes (2017)	Zapechelnyuk (2020)
Competition	Armstrong and Zhou (2022); Elliott et al. (2026); Li et al. (2026); Carnehl et al. (2026); Hwang et al. (2026)	This paper

Monopoly with an Exogenous Prior. Roesler and Szentes (2017) study buyer-optimal information before monopoly pricing when the buyer’s valuation is drawn from an exogenous distribution.² More generally, Kamenica and Gentzkow (2011) take the prior over states as given. Information therefore changes beliefs and subsequent decisions but not the distribution of the underlying state itself.

Monopoly with an Endogenous Prior. Zapechelnyuk (2020) studies optimal certification when a monopolistic producer privately chooses quality and sets a price. Certification therefore both informs consumers and affects incentives to provide quality. The certifier designs information to improve consumer welfare; in our monopoly benchmark, seller-optimal full disclosure can instead sustain efficient high quality.

More generally, Boleslavsky and Kim (2018) study persuasion with moral hazard in a principal-agent framework, where hidden effort determines the state distribution. Our endogenous prior is generated by the quality choices of competing sellers, so information also affects subsequent price competition.³

Competition with an Exogenous Prior. Armstrong and Zhou (2022) study information about consumer preferences under competition. Their seller-optimal

²Related work on information and monopoly pricing with exogenous consumer valuations includes Bergemann et al. (2015).

³Hörner and Lambert (2021) study a related dynamic problem in which ratings simultaneously provide information and incentives for costly effort, while Rosar (2017) studies test design under voluntary participation.

information amplifies perceived differentiation and relaxes price competition while preserving efficient allocation. [Elliott et al. \(2026\)](#) likewise study how information about heterogeneous consumer preferences affects competition. In our model, posterior differentiation additionally changes incentives to provide quality.

[Hwang et al. \(2026\)](#) study firms that choose prices and their own information provision. They show that equilibrium disclosure approaches full information as the number of firms grows. In our model, by contrast, a common seller-optimal information structure can favor opacity because it softens price competition and affects quality incentives.⁴

Most closely related are [Li et al. \(2026\)](#) and [Carnehl et al. \(2026\)](#), who study public information about exogenous qualities of vertically differentiated products. Revealing the quality ranking can maximize industry profits by creating perceived differentiation and relaxing price competition, whereas withholding ranking information can benefit consumers. Our model shares this differentiation mechanism, but sellers privately choose quality, so disclosure must also sustain quality incentives. Full disclosure is buyer-optimal in our model: it induces efficient, symmetric quality provision and zero seller profits. Sellers instead prefer partial disclosure that sustains quality randomization and creates differentiation in the buyer’s beliefs. The connection to the cited papers is especially close in our extension with correlated experiments (Section 5.1), where we show that a ranking experiment can be seller-optimal; however, unlike the truthful rankings in these papers, our optimal experiment sometimes ranks a low-quality product above a high-quality one to keep low quality profitable.

Competition with an Endogenous Prior. We characterize seller-optimal information under price competition when sellers privately choose quality. Seller-optimal information must simultaneously sustain quality incentives and create posterior differentiation. Under independent experiments, this requires partial information that keeps sellers indifferent between high and low quality.⁵

⁴Relatedly, [Gentzkow and Kamenica \(2017\)](#) show that competition among persuaders need not generally increase informativeness, while [Au and Kawai \(2020\)](#) obtain more aggressive disclosure as the number of competing senders increases.

⁵Related empirical work by [Vatter \(2025\)](#) estimates a model of quality investment and price competition; we provide an analytical characterization of seller-optimal information. [Boleslavsky and Cotton \(2015\)](#) study competition between schools with Bayesian persuasion, where schools choose both the grading policy and the quality of education. In their model, however, schools design the information themselves and do not compete in a product market through prices.

Certification, Disclosure, and Hidden Quality Our paper is also related to certification and quality disclosure. [Lizzeri \(1999\)](#) studies certification intermediaries that strategically choose what quality information to reveal, while [Albano and Lizzeri \(2001\)](#) endogenize quality provision and show that certification can strengthen quality incentives. More recent work takes an information design approach to the design and pricing of certification: [Celik and Strausz \(2026\)](#) and [Mäkimattila et al. \(2025\)](#) show how certifiers may limit the informativeness of experiments to maximize certification revenue. In our model, opacity instead helps soften competition between product sellers and incentivizes them to randomize over quality. We characterize information structures that maximize product sellers’ profits, abstracting from the contracting problem between certifiers and certification buyers that is central in the cited literature.

A related literature studies voluntary disclosure of privately known quality. [Board \(2009\)](#) shows that competition can prevent full unraveling, [Levin et al. \(2009\)](#) study costly disclosure under duopoly, and [Koessler and Renault \(2012\)](#) study the joint choice of price and verifiable product information.⁶ In much of this literature, quality is exogenous and firms themselves decide whether to reveal verifiable information.

Information and Product-Market Competition Finally, our paper relates to work on how information affects the intensity of competition. [Moscarini and Ottaviani \(2001\)](#) show how a buyer’s information about relative quality affects price competition, while [Ivanov \(2013\)](#) studies firms’ incentives to disclose product information under horizontal differentiation. In our model, sellers can benefit from coarse information because posterior differentiation softens price competition.

This mechanism is related to [Shaked and Sutton \(1982\)](#), who show how endogenous vertical product differentiation can relax price competition. Our model provides an informational counterpart: ex-ante identical sellers can differentiate through buyers’ posterior beliefs rather than deterministic product positions. Whereas [Shaked and Sutton](#)’s canonical differentiated-product outcome is asymmetric, we ask whether competition can be softened even in a symmetric environment, through the design of information rather than through asymmetric product choices. Because information also affects quality incentives, the opacity that relaxes competition induces sellers to randomize over quality.

⁶The classical unraveling logic goes back to, among others, [Grossman \(1981\)](#) and [Milgrom \(1981\)](#). [Dranove and Jin \(2010\)](#) survey the broader literature on quality disclosure and certification.

3. MODEL

Consider a market with a single buyer and either one seller or two sellers $i \in \{A, B\}$. Each seller has two goods available — one good of high quality and one good of low quality $q_i \in Q := \{0, 1\}$. The seller privately chooses which good she takes to the market. Her (opportunity) cost of selling this good, i.e., her valuation of the good, is $c \cdot q_i$ for some parameter $c \in (0, 1)$. The buyer has a unit demand and values a good with quality $q_i \in Q$ at q_i . The buyer receives information about the quality of the good through an exogenous experiment. After the experiment, the sellers compete in prices. The timing of the game is as follows.

Ex-Ante Stage At the ex-ante stage, sellers privately choose the quality of their good. The buyer’s ex-ante belief that the quality of seller i is high, $q_i = 1$, is $\mu_0 \in [0, 1]$, which is endogenously determined in equilibrium by sellers’ (mixed) quality strategies.⁷

Experiment Stage The buyer receives information about the quality of the goods through independent experiments. An experiment can be represented as a pair of the signal space S and a stochastic map $\pi : Q \rightarrow \Delta(S)$, where $\pi(s | q)$ denotes the probability of a signal realization s when the seller’s quality is q . The signals are observed publicly.⁸ We will work with the equivalent representation in terms of posterior beliefs: the experiment induces a post-experiment belief $\mu_i \in [0, 1]$ about the quality of seller i ’s good being high, $q_i = 1$. From the public signals, sellers can infer the buyer’s (equilibrium) post-experiment beliefs.

In equilibrium, an experiment gives rise to an *information structure*: conditional post-experiment belief distributions (F^0, F^1) , where F^q is the distribution function of μ_i conditional on the true quality $q \in \{0, 1\}$ of seller $i \in \{A, B\}$. Let $F(\mu_i) := \mu_0 F^1(\mu_i) + (1 - \mu_0) F^0(\mu_i)$ denote the unconditional distribution of post-experiment beliefs. Bayes plausibility requires (Kamenica and Gentzkow, 2011):

$$\int_0^1 \mu_i dF(\mu_i) = \mu_0.$$

⁷Throughout the paper, we will focus on symmetric equilibria, so that μ_0 captures the buyer’s ex-ante belief about each seller’s good.

⁸This assumption is particularly suitable in applications where buyers obtain information through certification processes.

Pricing Stage After observing the signals from the experiments, the sellers engage in Bertrand competition to sell their good by simultaneously choosing prices (p_A, p_B) .

Purchase Stage After observing prices, the buyer updates her beliefs about quality. Denote the resulting post-pricing beliefs by (μ_A^P, μ_B^P) . The buyer has a unit demand and decides whether to buy the good and, if so, from which seller. If the buyer does not buy the good, each player gets an ex-post payoff of zero. If she buys a good of quality q_i from seller i at price p_i , her ex-post payoff is

$$u^B = q_i - p_i,$$

and the profit of seller i is

$$\pi_i^S = p_i - cq_i,$$

while seller $j \neq i$ makes zero profit.

Equilibrium Concept The solution concept is perfect Bayesian equilibrium (PBE). We apply two refinements to the set of PBE.

Definition 1 (PBE with minimally informative prices). A profile of strategies and system of beliefs is a perfect Bayesian equilibrium with minimally informative prices if it satisfies the following:

- (i) (*Perfect Bayesian Equilibrium*) The strategy profile is sequentially rational given the system of beliefs, and the system of beliefs is derived from the strategy profile using Bayes' rule whenever possible.
- (ii) (*Cautiousness*) No seller chooses a price such that if the buyer were to buy her good at that price, the seller would make a negative profit. After any price $p_i < c$, the buyer's belief is $\mu_i^P = 0$.
- (iii) (*Minimally Informative Prices*) After any price $p_i \geq c$, the buyer's belief about seller i coincides with the post-experiment belief $\mu_i^P = \mu_i$.

Cautiousness requires that no seller chooses a price such that if the buyer were to buy her good at that price, the seller would make a negative profit. This criterion is standard in Bertrand games with heterogeneous costs or qualities; it rules out unreasonable equilibria in which a seller prices below her marginal cost since she expects she is not going to make a sale anyway. We call the criterion "cautiousness" following [Bergemann and Välimäki \(1996\)](#). By cautiousness, a high-quality seller never charges a price below c . Any such price, $p_i < c$, observed on the equilibrium path must therefore be attributed

to low quality, so that $\mu_i^P = 0$. We require the buyer's beliefs to respect cautiousness off the equilibrium path as well.⁹ Consequently, trade cannot occur at a positive price below c , and no seller can attract the buyer by undercutting c .

The requirement of *minimally informative prices* rules out some equilibria where the buyer's off-path beliefs dictate the equilibrium prices in an arbitrary way. The refinement allows us to concentrate on equilibria in which Bertrand competition works with minimal price signaling effects. For example, suppose that both sellers have the same post-experiment belief $\mu \in (c, 1)$. Without the refinement, both sellers charging μ can be supported as a pricing equilibrium by the belief that any undercutting seller has low quality. The refinement rules out such beliefs after prices at or above c , so that after any price $p_i \geq c$, the buyer's belief remains $\mu_i^P = \mu_i$.¹⁰

Discussion on Information Structures The model assumes that the information structure is exogenous, i.e., the experiment has been chosen prior to the game we analyze. In other words, we analyze the subgame after such an information structure has been designed. In applications, the experiment may be chosen by a certifier paid by sellers, a platform or marketmaker that charges sellers for participation. Thus, we focus on the seller-optimal information structure. Formally, we think of a designer who publicly commits to a symmetric and independent experiment before sellers choose quality. An information structure is seller-optimal if it maximizes each seller's ex-ante profit among such experiments, evaluated at the symmetric equilibrium it induces.

We further continue to focus on symmetric and independent information structures. Symmetry is motivated by the sellers' ex-ante identical positions. Independence, in turn, reflects an experiment like standardized certification, in which each product is evaluated on its own against a fixed benchmark, such as a safety rating, an inspection grade, or a laboratory test.

To further clarify the concept of information structures, we distinguish two experiments of special interest. The first one, a fully informative experiment,

⁹For a high-quality seller, any price below c is weakly dominated by $p_i = c$: it yields a loss if accepted and zero otherwise, whereas a low-quality seller gains from any accepted positive price. If $\mu_i = 1$, this is irrelevant, since a high-quality seller cannot gain from a price below c under any belief.

¹⁰The requirement of *minimally informative prices* follows the idea of [Koessler and Skreta \(2023\)](#). The issue is that the buyer may ignore the experiment and have arbitrary beliefs after an out-of-equilibrium price. [Koessler and Skreta \(2023\)](#) rule this out by saying that the experiment takes precedence. That is, if the sender acts out of equilibrium, but the experiment reveals the state (or more generally, reveals some event), then the receiver must not adopt a belief that is incompatible with what the experiment revealed.

perfectly reveals the quality.

Definition 2 (Fully informative experiment). An experiment is called *fully informative* if

$$F^0(\mu_i) = 1 \text{ for all } \mu_i \in [0, 1], \quad F^1(\mu_i) = \begin{cases} 0, & \text{for } \mu_i < 1, \\ 1, & \text{for } \mu_i = 1. \end{cases}$$

The natural counterpart of a fully informative experiment is a fully uninformative experiment. A fully uninformative experiment never changes the buyer's belief, so the game is as if there were no experiment at all.

Definition 3 (Fully uninformative experiment). An experiment is called fully uninformative if $F^0 \equiv F^1$, and F^0 and F^1 are degenerate distributions with unit mass on μ_0 .

We call an experiment partially informative or coarse if it is neither fully informative nor fully uninformative.

4. ANALYSIS

In this section, we first analyze the single seller model. Then we move to a model with two competing sellers.

4.1 SINGLE SELLER

Let us start by considering the situation of a single seller in the market, say, seller A . The seller sets her price after observing the signal from the experiment and is therefore aware of the buyer's realized post-experiment belief μ_A .

If $\mu_A \geq c$, the monopolist charges $p_A = \mu_A$ and sells the good. Higher prices are rejected by the buyer, while lower prices yield less profit. If $\mu_A < c$, any positive price that the buyer is willing to pay is below a high-quality seller's cost. Willingness to sell at such a price therefore reveals low quality, so that there is no trade at a positive price.¹¹

For later comparison, first hold the probability of high quality μ_0 fixed. The expected profit of the seller as a function of the posterior realization μ_A is $\mathbb{1}_{\mu_A \geq c} \mu_A (1 - c)$, as depicted in Figure 1. Given μ_0 , the expected profit of the seller is a convex combination of the seller's profit for different post-experiment

¹¹After post-experiment belief realization $\mu_A < c$, pricing in equilibrium can be as follows: the seller sets price $p_A = c$ regardless of her quality. The buyer's post-pricing belief is $\mu_A^P = \mu_A$ after any price $p_A \geq c$ and $\mu_A^P = 0$ after any price $p_A < c$. In equilibrium, the buyer always rejects trade.

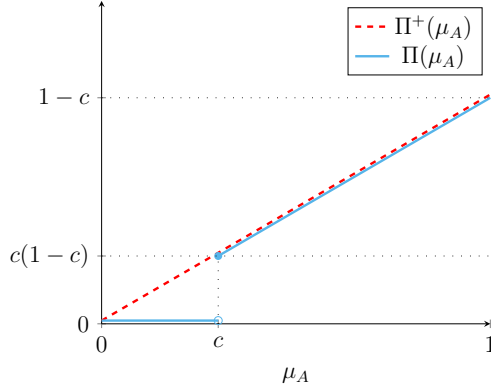


Figure 1: Expected profit $\Pi(\mu_A)$ as a function of posterior μ_A and the concave envelope $\Pi^+(\mu_A)$.

belief realizations. The optimal profit therefore lies on the concave envelope $\Pi^+(\mu_A) = \mu_A(1 - c)$, so that the maximal ex ante profit is $\mu_0(1 - c)$. A fully informative experiment attains this bound for any prior μ_0 ; other optimal experiments include a fully uninformative experiment when $\mu_0 \geq c$ and a distribution supported on $\{0, c\}$ when $\mu_0 < c$. The ex ante profit $\mu_0(1 - c)$ reaches its highest value when $\mu_0 = 1$, that is, when high quality is supplied with certainty.

We now ask which information structures (F^0, F^1) support high quality with certainty when the seller privately chooses quality. First, Bayes plausibility requires

$$F^1(\mu_A) = \begin{cases} 0, & \text{for } \mu_A < 1, \\ 1, & \text{for } \mu_A = 1. \end{cases} \quad (1)$$

Bayes plausibility does not restrict the selection of F^0 , the distribution of posterior beliefs under a deviation to low quality. However, to induce high quality in the equilibrium, the information structure must be such that the payoff from choosing high quality, $1 - c$, is at least as large as the expected payoff from deviating to low quality, that is,

$$1 - c \geq \int_c^1 \mu_A dF^0(\mu_A). \quad (2)$$

To explain the right-hand side, a deviating low-quality seller earns μ_A whenever $\mu_A \geq c$ and zero otherwise.

Proposition 1

With a single seller, the maximal profit $1 - c$ is attained when high quality is

produced and traded with certainty. Such an equilibrium is supported by an information structure if and only if (1) and (2) are satisfied.

One information structure that satisfies conditions (1) and (2) is the fully informative one. It perfectly reveals any deviation to low quality, making the deviation unprofitable.

4.2 TWO SELLERS

We now consider two competing sellers. A seller-optimal information structure must serve two purposes: soften price competition and sustain the quality choices on which sellers' profits depend. As before, we restrict attention to symmetric equilibria and symmetric, independent information structures.

Pricing follows from the buyer's posterior beliefs. As in the monopoly case, a positive price below c would reveal low quality and would not be accepted by the buyer. If $\mu_i > \mu_j \geq c$, seller i charges $c + \mu_i - \mu_j$ and makes the sale, while seller j charges c . The difference in beliefs allows seller i to charge a mark-up. If $\mu_i = \mu_j \geq c$, both sellers charge c and share the market. If $\mu_i \geq c > \mu_j$, seller i faces no effective competition and charges the monopoly price μ_i . If both posteriors are below c , there is no trade at a positive price. In each case, prices depend only on the public signals and not on quality, so that on the equilibrium path they convey no information and $\mu_i^P = \mu_i$, in line with the requirement of minimally informative prices.

Deterministic quality choices cannot sustain positive profits in a symmetric equilibrium. If both sellers choose high quality with certainty, Bayes plausibility implies that both posteriors equal one, and Bertrand competition drives prices to c . If both choose low quality with certainty, the buyer will not pay a positive price. Positive profits therefore require sellers to randomize quality, with a common probability $\mu_0 \in (0, 1)$ of choosing high quality. This requires indifference between the conditional profits π^1 and π^0 from choosing high and low quality, respectively.

Therefore, to find the seller-optimal information structure, we search for the posterior distribution F that maximizes ex ante profit per seller subject to this indifference condition. At the cutoff c , the buyer is indifferent between trading and not trading. To allow for both outcomes, we distinguish the non-trading posterior c^- from the trading posterior c^+ , both of which have numerical value

c.¹² The problem is

$$\begin{aligned} \max_{F \in \Delta([0, c^-] \sqcup [c^+, 1])} \quad & \mu_0 \pi^1 + (1 - \mu_0) \pi^0 \\ \text{subject to} \quad & \pi^1 = \pi^0, \quad \mu_0 = \int \mu dF(\mu) \in (0, 1). \end{aligned} \tag{3}$$

Here, the conditional profits are obtained from the pricing behavior described above, averaging over F^1 or F^0 for the seller and F for her rival. Choosing F thus determines both the information received by the buyer and the quality randomization that the information structure must sustain. In the discussion and figures below, c denotes the trading posterior unless otherwise stated.

To understand the problem better, let us consider two transformations of the posterior distribution. Each of them increases the ex-ante profit, but they may violate the indifference constraint. In a relaxed version of problem (3), in which this constraint is omitted, such transformations would always be profit-improving.

The first transformation is a mean-preserving spread of trading posteriors in $(c, 1)$ to $\{c, 1\}$. Figure 2 plots the expected sum of profits, $\Pi_{\mu_j}(\mu_i)$, as a function of one seller's posterior, holding the rival's posterior fixed. On the trading interval $[c, 1]$, this payoff is convex when the rival has posterior $\mu_j \geq c$ and linear otherwise. Spreading trading posteriors to the endpoints therefore weakly raises expected joint profits. The spread creates greater differentiation in the buyer's beliefs, allowing the seller with the higher posterior to charge a larger premium.

The second transformation raises all non-trading posterior mass to c^+ . Consider a realization (μ_i, μ_j) of the two sellers' posteriors. If at least one of the two is at or above c , the expected joint profit is unchanged: a rival whose posterior is exactly c charges c and leaves the buyer indifferent, and therefore constrains the other seller no more than a rival who does not trade at all. If both posteriors are below c , there is no trade before the transformation, whereas afterwards both sellers charge c and the joint profit is $c(1 - c) > 0$.

The appeal of posterior c is that it gives the buyer a belief that is sufficient to pay a positive price while preserving uncertainty about quality. We refer to it as *sufficient news*. Even when both sellers receive sufficient news and charge c , their expected joint profit is $c(1 - c) > 0$. A low-quality seller has a positive margin, but cannot gain by undercutting c : doing so would reveal her low quality. Information asymmetry thus prevents competition from driving low-quality profits to zero.

¹²We denote the set by $[0, c^-] \sqcup [c^+, 1]$, where $\sqcup$ is the disjoint union since posterior c appears in both sets but is distinct by its label.

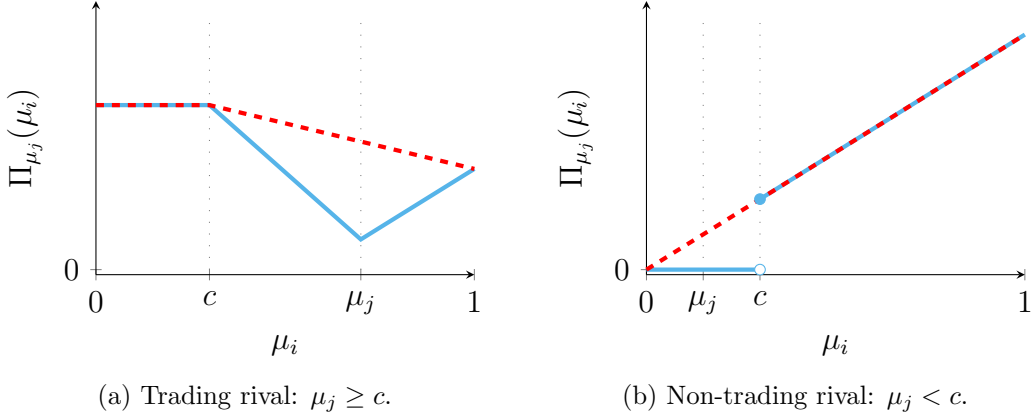


Figure 2: Expected joint profit as a function of one seller's posterior, holding the rival's posterior fixed. Blue: profits; red dashed: their concave envelopes.

These transformations imply that the optimum is supported on $\{c^+, 1\}$ in the relaxed version of the maximization problem without the indifference constraint. The following benchmark result verifies this and provides the optimal masses on these two posteriors, denoted by f_c and $f_1 = 1 - f_c$. This result can also be interpreted as the information structure that two sellers would choose if they could jointly commit to a common probability of high quality.

Lemma 1

Without the indifference constraint, the uniquely optimal posterior distribution in (3) is supported on $\{c^+, 1\}$, with

$$(f_c, f_1) = \left(\frac{1}{2-c}, \frac{1-c}{2-c} \right), \quad \mu_0 = \frac{1}{2-c}.$$

The maximal ex ante profit per seller is $\bar{\pi}(c) = (1-c)/[2(2-c)]$.

The optimal probability of high quality is the smallest prior at which the seller-optimal structure for a fixed μ_0 places no mass below c , and hence the smallest one at which trade occurs with probability one.¹³ Any lower μ_0 leaves non-trading mass in the market, whereas any higher μ_0 caps the probability of sufficient news at $f_c = (1 - \mu_0)/(1 - c) < \mu_0$ and hence reduces the associated low-quality rents.

The relaxed benchmark isolates the new restriction created by endogenous quality: sellers must be willing to randomize, and hence the two conditional

¹³For an arbitrary fixed probability of high quality μ_0 , the seller-optimal information structure is characterized in Lemma 2 in the Appendix.

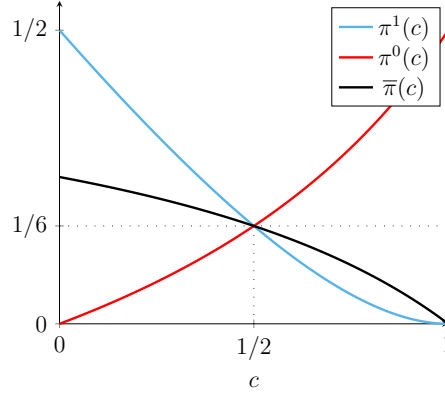


Figure 3: Profits in the solution to the relaxed problem that omits the indifference constraint. The indifference condition is satisfied only at $c = 1/2$.

profits must coincide. Under the information structure in Lemma 1,

$$\pi^1(c) = \frac{(1-c)^2}{2-c}, \quad \pi^0(c) = \frac{c}{2(2-c)}.$$

As Figure 3 illustrates, the relaxed solution satisfies the indifference condition of the full problem (3) only at $c = 1/2$. For $c < 1/2$, the relaxed information structure gives too strong an incentive to choose high quality; for $c > 1/2$, it gives too strong an incentive to choose low quality. Hence, except at $c = 1/2$, the relaxed upper bound cannot be attained if quality is chosen privately. If quality randomization and hence, positive profits are to be induced, a less informative structure when quality is cheap and a more informative one when it is expensive, is required. Hence not only the information structure but also the level of profits changes once quality is chosen privately: for every $c \neq 1/2$, sellers earn strictly less than $\bar{\pi}(c)$, and they would benefit from committing to a certain (optimal) quality randomization.

Our main result characterizes the seller-optimal information structure, taking into account that sellers must be willing to choose the quality distribution that generates the prior.

Theorem 1

For any $c \in (0, 1)$, there exists a seller-optimal symmetric information structure. Under any such structure F^ , sellers randomize between qualities and earn positive profit. Moreover:*

- (a) *If $c < 1/2$, there exists $\bar{\mu} \in (c, 1]$ such that F^* is supported on $[c^+, \bar{\mu}]$, has a positive atom at c^+ , and has a continuous full-support component with no atoms on $(c, \bar{\mu})$. An atom at the upper endpoint is possible only if $\bar{\mu} = 1$.*

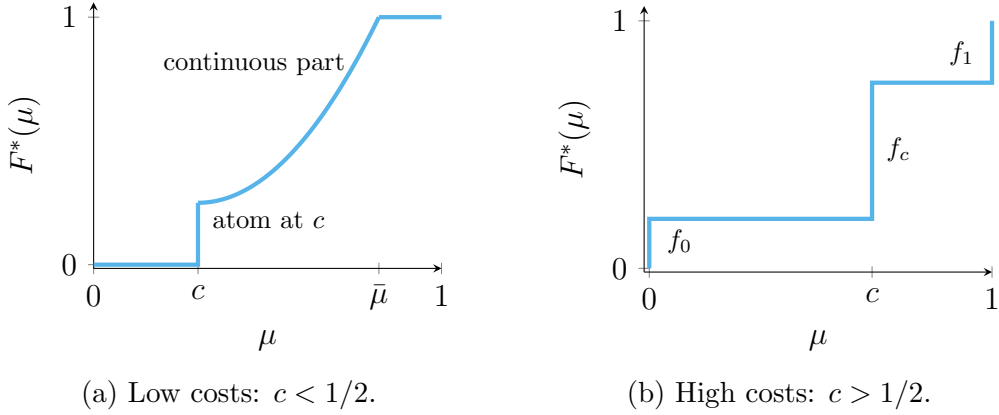


Figure 4: Optimal posterior distribution of Theorem 1.

- (b) If $c = 1/2$, F^* is given by Lemma 1.
- (c) If $c > 1/2$, F^* is supported on $\{0, c^+, 1\}$, with positive mass on each of these posteriors.

Figure 4 illustrates the two cases. With high costs, the optimal structure takes the **perfect news—sufficient news** form: the buyer either learns quality perfectly, corresponding to beliefs $\mu_i \in \{0, 1\}$, or she gets sufficient news $\mu_i = c$; sufficient in the sense that it is the lowest belief at which a seller can still make a sale, so that the buyer’s belief matches the lowest price a high-quality seller could ever ask for. With low costs, the optimal structure instead retains a continuous range of trading posteriors and potentially never perfectly reveals quality.

At the relaxed optimum of Lemma 1, indifference fails, and Figure 3 shows in which direction: for $c > 1/2$ the low-quality payoff is higher than the high-quality payoff, $\pi^0 > \pi^1$, whereas for $c < 1/2$ it is lower, $\pi^0 < \pi^1$. The proof establishes that the indifference constraint can indeed be relaxed to $\pi^1 \geq \pi^0$ when $c > 1/2$ and to $\pi^1 \leq \pi^0$ when $c < 1/2$, with equality at the optimum.

Recall the two transformations introduced above: (i) a mean-preserving spread of trading posteriors in $(c, 1)$ to $\{c, 1\}$, and (ii) a move of non-trading posterior mass to c^+ . Both weakly raise ex ante profit in the absence of indifference, but they move the relative profitability of the two qualities in opposite directions. Whether a transformation can be fully applied therefore depends on which inequality binds.

The first transformation, a mean-preserving spread of trading posteriors, raises π^1 and lowers π^0 through increasing informativeness. When $c > 1/2$, this relaxes the relevant inequality constraint $\pi^1 \geq \pi^0$, so all interior trading

posteriors in $(c, 1)$ are optimally spread to the endpoints. When $c < 1/2$, the indifference condition limits such a spread, and a continuous range of interior trading posteriors is left to keep low quality sufficiently profitable.

The second transformation, a move of non-trading posterior mass to the trading posterior c^+ , improves the low-quality profit π^0 relatively more than the high-quality profit π^1 . This is because even posterior $\mu_i = c$ is not high enough for a high-quality seller to ever make positive profit. When $c < 1/2$, this relaxes the relevant inequality constraint $\pi^1 \leq \pi^0$, so at the optimum, all posteriors are always in the trading region $[c, 1]$ and trade always occurs. When $c > 1/2$, the transformation pushes against the indifference constraint, and some non-trading mass is left: the information structure sometimes perfectly reveals bad news that strengthens incentives to produce high quality.

For all cost levels, $c \in (0, 1)$, the information structure of Theorem 1 is only partially informative and sellers randomize their quality. The partially informative experiment transforms quality uncertainty into differentiation in the buyer's beliefs. When the buyer receives different posterior beliefs about the two sellers, the seller with the higher posterior can charge a premium. When the buyer receives sufficient news about both sellers' goods, low-quality sellers still make positive rent. This is the central contrast with monopoly. A monopolist maximizes profit under full disclosure and when it chooses high quality with probability one. Relative to this, the seller-optimal information under competition distorts quality provision and reduces efficiency.

5. DISCUSSION

5.1 CORRELATED INFORMATION STRUCTURES

So far we have restricted the analysis to independent experiments, as when each product is evaluated on its own against a fixed benchmark. This assumption rules out experiments that directly compare the two products. We now allow all symmetric joint public experiments about (q_A, q_B) . Sellers continue to choose quality independently, with μ_0 denoting the probability of high quality.

Proposition 2

For every $c \in (0, 1)$, strictly positive profit can be attained by a symmetric joint public experiment. Every symmetric equilibrium with positive profit involves quality randomization with $\mu_0 \in (0, 1)$ and is supported by an experiment that is not fully revealing.

Allowing for correlated information structures therefore does not eliminate quality randomization or the need for opaque information. If both sellers always choose low quality, the buyer is not willing to buy the good at a positive

price. Conversely, if both sellers always choose high quality, price competition drives profits to zero. Positive profits require quality randomization, which full disclosure cannot sustain (see Section 4.2).

A simple “ranking experiment” can always sustain positive profits and is seller-optimal among all symmetric joint public experiments when quality is sufficiently costly. The experiment simply recommends one seller. When the products have the same quality, each seller is equally likely to be recommended. When they differ, the experiment recommends the high-quality seller with probability $(3 + c)/4$ and the low-quality seller with probability $(1 - c)/4$.

The ranking experiment supports a symmetric equilibrium in which each seller chooses high quality with probability $\mu_0 = 2c/(1 + c)$ and earns profit $c(1 - c)(3 - c)/(2 + 2c)$. The posterior belief about the recommended seller’s product is $c(3 - c)/(1 + c)$, and the posterior belief about the other seller’s product is c . The recommended seller makes the sale at a price equal to the buyer’s expected valuation for her product.

The experiment thus generates perceived differentiation at every realization, even when the products have the same quality. It favors the high-quality product sufficiently often to reward quality provision, but sometimes recommends the low-quality product to keep low quality profitable. Trade always occurs, and neither product’s quality is ever perfectly revealed.

Proposition 3

For $c \in [1/2, 1)$, the ranking experiment above is seller-optimal among all symmetric joint public experiments and their induced symmetric equilibria.

5.2 ASYMMETRIC EQUILIBRIA

Our previous analysis relies on selecting symmetric equilibria. Under full disclosure, asymmetric equilibria can instead reproduce the monopoly outcome.

Within asymmetric equilibria, there exists an asymmetric pure-strategy equilibrium in which one seller chooses high quality and the other chooses low quality. In this equilibrium, the high-quality seller acts as a monopolist, sets price $p = 1$, and earns profit $1 - c$, while the low-quality seller makes no sale and earns zero profit. Full revelation is seller-optimal in that case. The asymmetric equilibrium replicates the monopoly outcome from Section 4.1 and thus maximizes the sellers’ joint surplus. Given the continuation play pinned down by our refinements, the quality stage under full disclosure reduces to a 2×2 game:

		Seller 2	
		$q_2 = 1$	$q_2 = 0$
Seller 1	$q_1 = 1$	0, 0	$1 - c, 0$
	$q_1 = 0$	0, $1 - c$	0, 0

Under full disclosure, there are three pure-strategy equilibria, $\{(q_1=1, q_2=1), (q_1=1, q_2=0), (q_1=0, q_2=1)\}$, and a continuum of equilibria in which one seller chooses high quality with certainty and the other mixes. There is no equilibrium in which both sellers mix with strictly interior probabilities.¹⁴

If sellers could coordinate on an asymmetric equilibrium, full disclosure would maximize sellers' joint surplus. The symmetric equilibrium might nevertheless be the more compelling prediction. Observe that the asymmetric equilibria requires assigning different roles to sellers. We, however, assume that sellers are ex-ante identical. If sellers are also anonymous, i.e., there are no labels, no communication, and no repeated interaction, then sellers have no basis on which to coordinate on different roles, and a designer has no basis for treating them differently. A designer who could treat sellers asymmetrically, e.g., by assigning roles, could implement the monopoly outcome under full disclosure. We discuss this possibility for platforms in Section 5.3. Our comparison between monopoly and competition therefore applies when such coordination is unavailable.

Additionally, observe that in the payoff table low quality yields zero against every strategy of the rival, whereas high quality yields $1 - c > 0$ whenever the rival's quality is low. Hence in the reduced quality game holding fixed the continuation play pinned down by our refinements, low quality is weakly dominated. Every asymmetric and hybrid equilibrium therefore requires one seller to play a weakly dominated strategy, which is optimal for her only if she is *certain* that her rival chooses high quality. If sellers do not play weakly dominated quality choices, the unique outcome under full disclosure is $(q_1, q_2) = (1, 1)$, and profits are zero.

This criterion does not, however, affect the comparison between information structures, because it disciplines both regimes alike. Under the seller-optimal coarse structure, neither quality is weakly dominated. Since the rival's quality affects a seller's payoff only through the rival's signal distribution, payoffs in the reduced game are linear in the rival's mixing probability, and the indifference condition for interior $\mu_0 \in (0, 1)$ implies that the payoff differences

¹⁴To verify that these equilibria are also PBE in our model, recall that under full disclosure the experiment fully reveals quality, so that a deviation in quality is detected. A high-quality seller who deviates to low quality is revealed by the experiment, such that the buyer's posterior equals zero, and no sale occurs at a positive price. Hence no seller can profitably hide a deviation to low quality, and each of the three profiles is sequentially rational.

between the two qualities against the rival's pure strategies either both vanish or have opposite signs.¹⁵ In either case, no dominance relation exists, and the symmetric equilibrium with strictly positive profits survives the elimination that removes the asymmetric equilibria under full disclosure.

5.3 POLICY IMPLICATIONS

Information Design of Digital Platforms. Although stylized, the model speaks to information design on platforms. It abstracts from network effects, yet it isolates a simple mechanism through which platform information design may be driven as well. We focus throughout on seller-optimal information structures. This objective arises when a platform charges participation fees. Suppose the platform commits to an experiment and charges each seller a uniform fee before qualities are chosen. The fee is sunk at the quality and pricing stages and hence leaves the game we analyze unchanged. Given symmetry, the highest fee that both sellers are willing to pay is their ex-ante equilibrium profit, so that the platform's revenue equals the sellers' joint profit. The platform then chooses the information structure characterized in this paper: transparent with a single seller, coarse with competing sellers.¹⁶ With competing sellers, such intermediaries may prefer information structures that soften price competition, even when this leads to inefficient quality provision. This setting is particularly relevant for wholesale markets including agricultural markets and for online marketplaces, where platforms possess detailed information about products and control how it is disclosed to buyers.

The platform's control over market access, however, matters. The coarse structure is optimal for a platform that admits both sellers and treats them symmetrically. Suppose instead that the platform can restrict entry or assign roles. It could then admit a single seller under full disclosure and charge a fee of $1 - c$ or let a lottery determine which seller provides high quality in the asymmetric equilibrium and charge each seller $(1 - c)/2$. Either arrangement yields the monopoly surplus $1 - c$, which exceeds the sellers' joint profit under any symmetric information structure.¹⁷ Such a platform favors full transparency

¹⁵Our indifference condition implies that $\mu_0 [u(q_i = 1, q_j = 1) - u(q_i = 0, q_j = 1)] + (1 - \mu_0) [u(q_i = 1, q_j = 0) - u(q_i = 0, q_j = 0)] = 0$. Thus, either both differences are zero or have opposing signs.

¹⁶A commission on transaction prices aligns the platform with sellers in a similar way, but its revenue then depends on expected transaction prices and revenue rather than on profits, and the commission raises sellers' effective cost of high quality. The incentive to differentiate the buyer's beliefs remains, but the optimal information structure would differ from the one characterized here.

¹⁷By Lemma 1, the joint profit is at most $(1 - c)/(2 - c) < 1 - c$.

and efficient quality provision, but extracts the entire surplus, leaving consumers with zero surplus. Our results therefore speak to open marketplaces in which the platform cannot or may not discriminate between sellers, for instance because buyers value variety or because of non-discrimination obligations.

Mandatory Disclosure. A natural regulatory response to opaque information structures is to mandate full disclosure. In the symmetric equilibrium, this restores efficient quality provision: both sellers choose high quality, Bertrand competition drives profits to zero, and consumers obtain the entire surplus $1 - c$. Relative to the seller-optimal coarse structure, mandatory disclosure thus raises both total surplus and consumer surplus.

This conclusion, however, relies on the absence of coordination. Under full disclosure, there also exist asymmetric equilibria in which one seller chooses high quality and acts as a monopolist, while the other chooses low quality and makes no sale. Allocation is then efficient, but consumer surplus is zero. If an intermediary can coordinate sellers on such roles or restrict entry, mandatory disclosure improves quality provision but leaves the entire surplus to sellers and the intermediary. Under the coarse information structure, by contrast, consumers obtain positive surplus, as competition keeps prices below the monopoly level. A regulator focused solely on quality provision may therefore harm consumers if mandatory disclosure is combined with coordinated or restricted market access.

Certification. In many markets, buyers' information about quality is produced by certifiers, e.g., safety ratings, inspection grades, or laboratory seals. Our analysis offers one explanation for why certification often falls short of guaranteeing product quality. When certification is financed by the industry it evaluates, its design might be similar to the seller-optimal structure we characterize; that is, coarse. For high costs, this structure resembles threshold certification with occasional revelation. The atom at the sufficient-news belief c corresponds to a pass mark calibrated to the lowest belief at which trade occurs, while the mass at zero corresponds to withholding or revoking the seal. Because sellers randomize over quality in equilibrium, a certified product need not be a high-quality product: incomplete certification arises endogenously, as it relaxes price competition among the certified. Importantly, this force is specific to competition as a certifier serving a monopolist would prefer to disclose fully. Our perspective thereby complements [Lizzeri \(1999\)](#), in which a profit-maximizing certifier reveals minimal information in order to extract fees, and [Albano and Lizzeri \(2001\)](#), in which certification under quality moral hazard with a single seller improves incentives to provide high quality goods.

6. CONCLUSION

This paper studies how the interaction between information and market structure shapes prices and quality provision. Focusing on information structures that maximize sellers' profits, we find a sharp contrast between monopoly and competition. A monopolist profits most from full disclosure, which induces the efficient provision of high quality. Competing sellers, by contrast, profit most from coarse information structures that keep them exactly indifferent between qualities and thereby induce randomization in quality choice. The form of optimal disclosure depends on the cost of quality. When high quality is expensive, the optimal structure combines perfectly revealing signals with sufficient news. When high quality is inexpensive, it retains a continuous range of noisy posterior beliefs and may never reveal quality perfectly, while ensuring that trade always occurs. In each case, quality provision is inefficient when information disclosure is aligned with competing sellers' interests.

Our results have several limitations. For example, the comparison with monopoly relies on equilibrium selection: under full disclosure there also exist asymmetric equilibria that replicate the monopoly outcome, in which case competition need not reduce quality. Additionally, we restrict attention to symmetric experiments. Future extensions of the model could allow for an endogenous designer or intermediary and the relationship between the designer's incentives of information provision and competition in the marketplace.

APPENDIX: PROOFS

PROOF OF LEMMA 1

To prove Lemma 1, we first solve for the seller-optimal information structure with exogenous quality.

Lemma 2

Assume that there are two sellers and that quality is high with probability μ_0 , independently across sellers. In any optimal information structure, $\mu_i \in \{0, c, 1\}$ with probability one and

(i) *if $\mu_0 \leq \frac{1}{2-c}$, then*

$$(f_0, f_c, f_1) = (1 - \mu_0(2 - c), \mu_0, (1 - c)\mu_0), \text{ and}$$

(ii) *if $\mu_0 > \frac{1}{2-c}$, then*

$$(f_0, f_c, f_1) = \left(0, \frac{1 - \mu_0}{1 - c}, \frac{\mu_0 - c}{1 - c}\right).$$

Proof. Take arbitrary unconditional cumulative distribution function of posteriors F and say that μ_A, μ_B are each (independently) distributed according to F .

Given posterior realization μ_j for seller j , let us write the expected sum of profits for the two sellers as a function of posterior realization μ_i for seller $i \neq j$:

$$\Pi_{\mu_j}(\mu_i) = \mathbb{1}_{\max\{\mu_A, \mu_B\} \geq c} \max\{\mu_A, \mu_B\}(1 - c) - \max\{0, \min\{\mu_A, \mu_B\} - c\}.$$

For any fixed μ_j , $\Pi_{\mu_j}(\mu_i)$ is convex in μ_i on interval $[0, c]$, as it is constant on $[0, c)$ and $\Pi_{\mu_j}(c) \geq \lim_{\mu_i \rightarrow c^-} \Pi_{\mu_j}(\mu_i)$. Furthermore, $\Pi_{\mu_j}(\mu_i)$ is convex on interval $[c, 1]$, as it is linearly decreasing on $[c, \max\{c, \mu_j\}]$ and linearly increasing on $[\max\{c, \mu_j\}, 1]$. Then also $\mathbb{E}_{\mu_j \sim F}[\Pi_{\mu_j}(\mu_i)]$ is convex in μ_i on intervals $[0, c]$ and $[c, 1]$ as a (weighted) sum of convex functions is also convex.

Consequently, following [Kamenica and Gentzkow \(2011\)](#), a mean-preserving spread of posteriors $\mu_i \in (0, c)$ (performed independently of μ_j) to the end points of the interval weakly increases the sellers' total expected profit. Similarly, a mean-preserving spread of posteriors $\mu_i \in (c, 1)$ to the end points of the interval weakly increases the sellers' total profit. Furthermore, if the original information structure F assigns a positive probability to posterior realizations

on open intervals $(0, c)$ and $(c, 1)$, these mean-preserving spreads strictly increase the sellers' total profit. Analogous mean-preserving spreads of posteriors μ_j also weakly increase the sellers' total expected profit. This procedure produces a new (symmetric, independent and mean-preserving) information structure with support $\{0, c, 1\}$ that produces a weakly greater profit for the sellers than the original information structure (strictly greater if the original information structure F assigns a positive probability to posterior realizations on open intervals $(0, c)$ and $(c, 1)$). Hence, the sellers' profit is maximized with an information structure such that $\mu_i \in \{0, c, 1\}$ with probability 1 for both $i \in \{A, B\}$.

Letting $f_{\hat{\mu}} := \Pr[\mu_i = \hat{\mu}]$ denote the ex-ante probability that the buyer's posterior about the good of any seller i is $\hat{\mu}$, the problem of finding the seller-optimal information structure has now been reduced to one of finding f_0, f_c and f_1 to maximize the sellers' expected profit:

$$\max_{f_0, f_c, f_1 \in [0, 1]} (f_c^2 + 2f_0f_c)c(1 - c) + 2f_1(1 - f_1)(1 - c) \quad (4)$$

$$\text{s.t. } \mu_0 = f_c c + f_1 \quad (5)$$

$$f_0 + f_c + f_1 = 1 \quad (6)$$

Note that if $\mu_A = \mu_B = 0$ or $\mu_A = \mu_B = 1$, the sellers do not make profit. They make positive profit if $\max\{\mu_A, \mu_B\} = c$, which happens with probability $f_c^2 + 2f_0f_c$ and in which case the expected profit is $c(1 - c)$. They also make positive profit if $\mu_i = 1$ and $\mu_j < 1$ for $i, j \in \{A, B\}$, $i \neq j$, which happens with probability $2f_1(1 - f_1)$ and in which case the expected profit is $1 - c$. Therefore, the sellers' total expected profit given f_0, f_c and f_1 is given by (4). Constraint (5) is a Bayes plausibility constraint and constraint (6) is a feasibility constraint; they can be plugged to the objective to make f_c the only decision variable:

$$\max_{f_c \in \mathbb{R}} (1 - c) \left((2\mu_0 - f_c)cf_c + 2(1 - \mu_0)\mu_0 \right) \quad (7)$$

$$\text{s.t. } f_c \geq 0 \quad (8)$$

$$f_c \leq \frac{\mu_0}{c}, \quad (9)$$

$$f_c \leq \frac{1 - \mu_0}{1 - c} \quad (10)$$

Here, the constraints ensure that the solution satisfies $f_0, f_c, f_1 \in [0, 1]$. If the constraints are ignored, the unique solution to the problem is $f_c = \mu_0$. This solution satisfies the constraints if $\mu_0 \leq \frac{1}{2-c}$; otherwise it is uniquely optimal to set $f_c = \frac{1-\mu_0}{1-c}$. Given f_c , we can then find f_0 and f_1 from (5) and (6) to obtain Proposition 2. $\square$

It remains to optimize the resulting joint profit over μ_0 . Given the information structure of Lemma 2, the joint expected profit as a function of μ_0 is

$$\Pi(\mu_0) = \begin{cases} (1-c)\mu_0(2-(2-c)\mu_0), & \mu_0 \leq \frac{1}{2-c}, \\ \frac{(1-\mu_0)((2-c)\mu_0-c)}{1-c}, & \mu_0 > \frac{1}{2-c}. \end{cases} \quad (11)$$

Both expressions equal $(1-c)/(2-c)$ at $\mu_0 = 1/(2-c)$, so Π is continuous. For $\mu_0 < 1/(2-c)$, $\Pi'(\mu_0) = 2(1-c)(1-(2-c)\mu_0) > 0$. For $\mu_0 > 1/(2-c)$, $\Pi'(\mu_0) = \frac{2-2(2-c)\mu_0}{1-c} < 0$. Therefore Π is maximized at $\mu_0^* = 1/(2-c)$.

Substituting μ_0^* into (11) gives the maximal joint profit. Substituting μ_0^* into the formulas of Lemma 2 gives the posterior masses in Lemma 1. $\square$

PROOF OF THEOREM 1

Step I: Existence.

Lemma 3 (Existence)

Fix $c \in (0, 1)$. Consider symmetric equilibria generated by symmetric and independent information structures, allowing continuation play to prescribe either trade or no trade, depending on the observed signal label, when the relevant posterior equals c . Within this class, there exists a seller-optimal information structure and equilibrium. The optimum induces interior randomization over quality and strictly positive seller profits.

Proof. At every posterior different from c , the buyer's optimal continuation action is uniquely determined by the posterior: there is no trade below c and trade above c . At the cutoff c , however, the buyer is indifferent between trading and not trading. Hence, the posterior alone is not sufficient to describe an equilibrium outcome. In particular, two distinct signal realizations may induce the same posterior c but lead to different equilibrium best-response selections by the buyer, and therefore to different seller payoffs.

To retain this payoff-relevant continuation behavior in the reduced form, we split the cutoff posterior into two labelled copies. We write c^- for posterior c followed by no trade and c^+ for posterior c followed by trade. Accordingly, define the split posterior space

$$\mathcal{X}_c := [0, c^-] \sqcup [c^+, 1],$$

where

$$I_c := [0, c^-] \quad \text{and} \quad A_c := [c^+, 1]$$

are respectively the no-trade and trade components. We retain the usual topology on each interval and treat the two intervals as separate components.

Since both components are compact, $\mathcal{X}_c$ is compact and metrizable, and I_c and A_c are clopen in $\mathcal{X}_c$.

Let

$$\mu : \mathcal{X}_c \longrightarrow [0, 1]$$

denote the underlying numerical posterior. Thus, it holds that $\mu(z) = z$ except from

$$\mu(c^-) = \mu(c^+) = c.$$

Both c^- and c^+ have the same posterior belief c and differ only in the buyer's equilibrium continuation action.

Given a pair of labelled posterior realizations $(z, z') \in \mathcal{X}_c^2$, the buyer's equilibrium strategy is as follows. If $z, z' \in I_c$, the buyer does not trade. Otherwise, among the sellers whose realizations belong to A_c , the buyer trades with a seller having the highest numerical posterior, with symmetric tie-breaking if both active sellers have the same posterior.

This strategy is sequentially rational. If both realizations lie in I_c , all available numerical posteriors are weakly below c , so no trade is a best response. If an active seller has posterior strictly above c , trading with an active seller having the highest posterior is optimal. Finally, if the highest active posterior equals c , the buyer is indifferent between trading and not trading, so the continuation prescribed by the label c^+ is also a best response. Thus the split space does not give the information designer control over the buyer's action; it merely records which of the buyer's payoff-equivalent best responses is selected in equilibrium at the cutoff.

A symmetric and independent information design over $\mathcal{X}_c$ is represented in reduced form by a probability distribution

$$F \in \Delta(\mathcal{X}_c).$$

The distribution F is the common unconditional distribution of each seller's labelled posterior-continuation outcome. The induced prior probability of high quality is

$$\mu_0(F) := \int_{\mathcal{X}_c} \mu(z) \, dF(z). \quad (12)$$

Let k^1 and k^0 assign to each pair (z, z') of labelled posterior-continuation outcomes its contribution to the weighted profit from high and low quality, respectively. The functions are symmetrized across the two sellers. If $(z, z') \in [0, c^-] \times [0, c^-]$, no seller trades and they make zero profits. If $z \in [c^+, 1]$ and $z' \in [0, c^-]$ (and vice versa), profits are

$$k^1(z, z') = \frac{1}{2} \mu(z) (\mu(z) - c)$$

and

$$k^0(z, z') = \frac{1}{2}(1 - \mu(z))\mu(z),$$

If $(z, z') \in [c^+, 1] \times [c^+, 1]$, they are

$$k^1(z, z') = \frac{1}{2} \max\{\mu(z), \mu(z')\} |\mu(z) - \mu(z')|$$

and

$$k^0(z, z') = \frac{1}{2}(1 - \max\{\mu(z), \mu(z')\}) (|\mu(z) - \mu(z')| + c).$$

The weighted quality dependent profits are therefore

$$\mu_0(F)\pi^1(F) := \int_{\mathcal{X}_c} \int_{\mathcal{X}_c} k^1(z, z') dF(z') dF(z) \quad (13)$$

and

$$(1 - \mu_0(F))\pi^0(F) := \int_{\mathcal{X}_c} \int_{\mathcal{X}_c} k^0(z, z') dF(z') dF(z). \quad (14)$$

The seller's ex-ante expected profit is

$$\mu_0(F)\pi^1(F) + (1 - \mu_0(F))\pi^0(F).$$

Seller's Problem. The seller-optimal information design problem is therefore

$$\begin{aligned} & \sup_{F \in \Delta(\mathcal{X}_c)} \mu_0(F)\pi^1(F) + (1 - \mu_0(F))\pi^0(F), \\ & \text{subject to } \pi^1(F) = \pi^0(F), \text{ and } \mu_0(F) \in (0, 1). \end{aligned} \quad (15)$$

Denote the feasible set by

$$\Phi := \{F \in \Delta(\mathcal{X}_c) : \pi^1(F) = \pi^0(F), \mu_0(F) \in (0, 1)\}. \quad (16)$$

The constraint $\pi^1(F) = \pi^0(F)$ is stated in terms of the conditional profits, which are obtained by dividing the weighted profits by $\mu_0(F)$ and $1 - \mu_0(F)$ and are therefore undefined when $\mu_0(F) \in \{0, 1\}$. To obtain a closed superset of Φ , we restate the constraint using the weighted profits, which are defined for every $F \in \Delta(\mathcal{X}_c)$:

$$\widehat{\Phi} := \{F \in \Delta(\mathcal{X}_c) : (1 - \mu_0(F))[\mu_0(F)\pi^1(F)] = \mu_0(F)[(1 - \mu_0(F))\pi^0(F)]\},$$

Whenever $\mu_0(F) \in (0, 1)$, dividing by $\mu_0(F)(1 - \mu_0(F)) > 0$ shows that the defining equation is equivalent to $\pi^1(F) = \pi^0(F)$.

Application of Weierstrass Theorem. Because the four components of $\mathcal{X}_c^2$ are clopen, k^1 and k^0 are bounded and continuous. It follows that $\mu_0(F)$, $\mu_0(F)\pi^1(F)$, and $(1 - \mu_0(F))\pi^0(F)$ are continuous as well. Both sides of the equation defining $\widehat{\Phi}$ are products of these continuous functions, so $\widehat{\Phi}$ is closed in the compact space $\Delta(\mathcal{X}_c)$ and is therefore compact. The ex-ante profit

$$\mu_0(F)\pi^1(F) + (1 - \mu_0(F))\pi^0(F) \quad (17)$$

is also continuous. By Weierstrass' theorem, it attains a maximum on $\widehat{\Phi}$ at some F^* .

It remains to show that the maximum is strictly positive. We construct, for every $c \in (0, 1)$, a perfect-news-sufficient-news posterior distribution under which sellers randomize over quality and earn strictly positive profits. The distribution has support $\{0, c^+, 1\}$. Let

$$f_0 := F(\{0\}), \quad f_c := F(\{c^+\}), \quad f_1 := F(\{1\}).$$

Thus f_c denotes the probability of trade at the numerical posterior c . Bayes plausibility requires

$$\mu_0 = cf_c + f_1. \quad (\text{BP})$$

The expected profit from choosing high quality is

$$\pi^1 = \frac{f_1}{\mu_0}(f_0 + f_c)(1 - c),$$

whereas the expected profit from choosing low quality is

$$\pi^0 = \frac{1 - c}{1 - \mu_0} [(f_0 + f_c)^2 - f_0^2] \frac{c}{2}.$$

For sellers to randomize over quality, the two profits must be equal:

$$\pi^1(F) = \pi^0(F). \quad (\text{I})$$

The posterior probabilities must also satisfy

$$f_0 + f_c + f_1 = 1. \quad (\text{F})$$

Fix $\mu_0 = c$ and define

$$f_c^* := (1 - c) \left(1 + \frac{\sqrt{1 + c^2} - 1}{c} \right),$$

together with

$$f_1^* := c(1 - f_c^*)$$

and

$$f_0^* := (1 - c)(1 - f_c^*).$$

The value f_c^* is the unique solution in $(0, 1)$ to

$$c(f_c^*)^2 = 2(1 - c)^2(1 - f_c^*). \quad (18)$$

By construction, (BP) and (F) hold. Then the seller's expected profit is

$$\frac{1}{c}(1 - c)^2 \left(c + \sqrt{1 + c^2} \right) \left(1 - \sqrt{1 + c^2} - c(1 - c - \sqrt{1 + c^2}) \right) > 0.$$

If $\mu_0(F) = 0$, Bayes plausibility implies that $\mu(z) = 0$ almost surely, and profits are zero. If $\mu_0(F) = 1$, then $\mu(z) = 1$ almost surely, and symmetric Bertrand competition again yields zero profits. Since the maximum is strictly positive, it follows that

$$\mu_0(F^*) \in (0, 1), \quad (19)$$

and

$$\pi^1(F^*) = \pi^0(F^*). \quad (20)$$

Hence F^* solves the original interior-quality problem.

Finally, $F^* = \mu_0^* F^1 + (1 - \mu_0^*) F^0$ is a Bayes-plausible experiment. Since the indifference condition is fulfilled and $\mu_0(F) \in (0, 1)$, sellers randomize over quality in equilibrium and earn strictly positive profits. $\square$

Step II: Characterization. Previously, we have shown that there exists a seller-optimal information structure and it solves

$$\begin{aligned} \max_{F \in \Delta(\mathcal{X}_c)} \quad & \mu_0(F) \pi^1(F) + (1 - \mu_0(F)) \pi^0(F) \\ \text{subject to} \quad & \pi^1(F) = \pi^0(F), \mu_0(F) \in (0, 1). \end{aligned} \quad (21)$$

Throughout the characterization, we use $\mu_i \in \mathcal{X}_c$ to denote seller i 's labelled posterior-continuation outcome. With a slight abuse of notation, μ_i denotes the underlying numerical posterior associated with that outcome. Thus c^- and c^+ are distinct elements of $\mathcal{X}_c$, but both have numerical posterior c . In intervals and other set expressions, we continue to use those labels.

Now, the induced prior is

$$\mu_0 = \int_{[0, c^-]} \mu_i dF(\mu_i) + \int_{[c^+, 1]} \mu_i dF(\mu_i). \quad (22)$$

For $\mu_0(F) \in (0, 1)$, the weighted expected profit conditional on high quality is

$$\mu_0\pi^1 = \int_{[c^+, 1]} \mu_i \left[F([0, c^-])(\mu_i - c) + \int_{[c^+, \mu_i)} (\mu_i - \mu_j) dF(\mu_j) \right] dF(\mu_i), \quad (23)$$

and on low quality, respectively, is

$$(1 - \mu_0)\pi^0 = \int_{[c^+, 1]} (1 - \mu_i) \left[F([0, c^-])\mu_i + \int_{[c^+, \mu_i)} (\mu_i - \mu_j + c) dF(\mu_j) + \frac{c}{2} F(\{\mu_i\}) \right] dF(\mu_i). \quad (24)$$

We prove Theorem 1 through multiple lemmata, which will simplify the characterization of the seller-optimal information structure. The first lemma implies that we can replace all posterior mass on $[0, c^-]$ with one atom $a \in [0, c^-]$ without changing the profit and incentives.

Lemma 4

For every $F \in \Delta([0, c^-] \cup [c^+, 1])$, there exist $a \in [0, c^-]$ and $F_a \in \Delta(\{a\} \cup [c^+, 1])$ that induce the same prior, the same conditional profits, and hence the same ex ante profit.

Proof. Observe that the restriction of F to $[0, c^-]$ affects $\mu_0\pi^1$ in Equation 23 and $(1 - \mu_0)\pi^0$ in Equation 24 only through its probability mass on $[0, c^-]$, and it affects the prior $\mu_0(F)$ only through its first moment (see Equation 22).

Hence, we can replace any feasible distribution $F \in \Delta([0, c^-] \cup [c^+, 1])$ that has some mass on $[0, c^-]$ with another distribution $F_a \in \Delta(\{a\} \cup [c^+, 1])$ such that $F_a(\mu) = F(\mu)$ for all $\mu \in [c^+, 1]$ and $a = \frac{\int_{[0, c^-]} \mu dF(\mu)}{F([0, c^-])}$, where $a = 0$ and $F_a = F$ if there is no mass on $[0, c^-]$. Hence F_a concentrates all posterior mass of F on $[0, c^-]$ to one point a . By construction, F_a and F induce the same prior. They also induce the same weighted conditional profits $\mu_0\pi^1$ and $(1 - \mu_0)\pi^0$. $\square$

Next, we establish that a mean-preserving spread weakly raises the sellers' ex-ante profits.

Lemma 5

A mean-preserving spread of all posteriors $\mu \in (c, 1)$ to $\{c^+, 1\}$ weakly raises π^1 , weakly lowers π^0 , and weakly raises the ex-ante profit $\mu_0\pi^1 + (1 - \mu_0)\pi^0$.

Proof. Replace each posterior $\mu \in (c, 1)$ with a lottery under which outcome 1 is realized with probability

$$\frac{\mu - c}{1 - c},$$

and outcome c^+ is realized with probability

$$\frac{1 - \mu}{1 - c}.$$

This lottery is mean preserving because the numerical posterior associated with c^+ is c . Apply this transformation independently to the posteriors of both sellers. The transformation preserves the prior μ_0 .

Consider first two outcomes in the trade component $[c^+, 1]$ with numerical posterior values $\mu' \geq \mu'' \geq c$. We study the contribution of the case where the posterior realizations of the two sellers are (μ', μ'') to the expected profit of a seller (i.e., summing over the case where that seller has posterior μ' against μ'' and the case where the seller has posterior μ'' against μ').

Before the spread, the contributions to $\mu_0\pi^1$ and $(1-\mu_0)\pi^0$ are, respectively,

$$\mu'(\mu' - \mu'') \quad \text{and} \quad (1 - \mu')(\mu' - \mu'' + c).$$

After both posteriors have been spread over $\{c^+, 1\}$, the contribution to $\mu_0\pi^1$ becomes

$$\frac{(\mu' - c)(1 - \mu'') + (1 - \mu')(\mu'' - c)}{1 - c}.$$

Subtracting the original contribution gives

$$\frac{(1 - \mu')\{(1 - c)(\mu' - c) + (1 + c)(\mu'' - c)\}}{1 - c} \geq 0. \quad (25)$$

For the low-quality payoff, the transformed pair contributes positively only when both outcomes become c^+ . Its new contribution is therefore

$$\frac{c(1 - \mu')(1 - \mu'')}{1 - c}.$$

Subtracting the original contribution gives

$$-\frac{(1 - \mu')[(1 - c)(\mu' - \mu'') + c(\mu'' - c)]}{1 - c} \leq 0. \quad (26)$$

Adding (25) and (26) gives the change in ex ante profit:

$$\frac{(1 - \mu')(2 - c)(\mu'' - c)}{1 - c} \geq 0. \quad (27)$$

Suppose instead that the pair of numerical posteriors (μ', μ'') is such that $\mu' \geq c > \mu''$, i.e., one lies within the no-trade component, whereas the other

lies in the trade component. Before the spread, their contributions to the weighted high- and low-quality profits are

$$\mu'(\mu' - c)$$

and

$$(1 - \mu')\mu',$$

respectively. After the spread, these contributions become

$$\mu' - c$$

and

$$c(1 - \mu'),$$

respectively. Hence the high-quality contribution changes by

$$(1 - \mu')(\mu' - c) \geq 0,$$

while the low-quality contribution changes by

$$-(1 - \mu')(\mu' - c) \leq 0.$$

Their sum is unchanged.

Pairs for which both outcomes lie in the no-trade component are unaffected. It follows that the transformation weakly raises $\mu_0\pi^1$, weakly lowers $(1 - \mu_0)\pi^0$, and weakly raises their sum. Since the transformation preserves μ_0 , it also weakly raises π^1 and weakly lowers π^0 . □

Now, we turn to characterizing the seller-optimal information structure. We split the characterization into a high-cost, $c \geq 1/2$, and a low-cost, $c < 1/2$, part.

Lemma 6

If $c > 1/2$, every seller-optimal information structure is supported on $\{0, c^+, 1\}$.

Proof. Fix $c > 1/2$ and let F be a seller-optimal information structure satisfying the indifference condition. We construct, through a sequence of operations another distribution that satisfies the indifference condition and has mass only on posteriors $\{0, c^+, 1\}$, which yields weakly higher ex-ante profit. Finally, we show that the probability mass on each of these posteriors is optimally positive.

First, use Lemma 4 to collapse the lower tail $[0, c^-]$ to its conditional mean a and apply the mean-preserving spread of Lemma 5 to $(c, 1)$. The resulting

distribution is supported on $\{a, c^+, 1\}$, yields weakly higher profit, and satisfies $\pi^1 \geq \pi^0$. Denote the masses on a , c^+ , and 1 by (f_a, f_c, f_1) .

We now perform two operations: we move the lower atom a to zero, and move some mass from a to c , holding f_1 fixed, while preserving $\pi^1 - \pi^0$. This leaves the weighted high-quality payoff $\mu_0 \pi^1 = (1 - c)f_1(1 - f_1)$ unchanged and raises the weighted low-quality payoff $(1 - \mu_0)\pi^0 = (1 - c)f_c(f_a + \frac{f_c}{2})$. It therefore raises ex ante profit and lowers $\pi^1 - \pi^0$. The latter change can be offset by lowering a : this lowers μ_0 and hence raises $\pi^1 - \pi^0$, without affecting either weighted payoff or the ex-ante profit. By continuity, the two changes can be combined so that $\pi^1 - \pi^0$ remains constant. This adjustment can be continued until either $a = 0$, in which case the support is contained in $\{0, c^+, 1\}$, or the mass at a is exhausted, in which case the support is contained in $\{c^+, 1\}$.

Suppose first that mass at 0 remains. Move mass t from 0 to c , keeping f_1 fixed. This leaves

$$\mu_0 \pi^1 = (1 - c)f_1(1 - f_1)$$

unchanged, whereas

$$\frac{d}{dt}[(1 - \mu_0)\pi^0] = (1 - c)c(f_0 - t) \geq 0.$$

The prior also rises, so this operation must increase π^0 and decrease π^1 , hence we move closer to the indifference $\pi^1 = \pi^0$ while raising profit. Continue this operation until either the mass on 0 is exhausted or the indifference is reached. If indifference is reached first, we are done.

Finally, suppose that the mass below c is exhausted (and hence the support is $\{c^+, 1\}$) while $\pi^1 > \pi^0$. Note that this implies that f_c is below the mass $\hat{f}_c$ under which indifference would hold,

$$f_c < \hat{f}_c := \frac{2 - 3c}{(1 - c)(2 - c)}.$$

This can arise only when $c < 2/3$, since $\hat{f}_c \leq 0$ for $c \geq 2/3$. The ex-ante profit is

$$\mu_0 \pi^1 + (1 - \mu_0)\pi^0 = (1 - c)f_c \left[1 - \left(1 - \frac{c}{2} \right) f_c \right]$$

and its derivative with respect to f_c is

$$(1 - c)[1 - (2 - c)f_c]$$

This derivative is non-negative as long as $f_c < \hat{f}_c$ and $1/2 \leq c < 2/3$, hence moving mass from 1 to c until $f_c = \hat{f}_c$ restores indifference and raises profit.

We have thus obtained an indifferent distribution with probability mass only on posteriors $\{0, c^+, 1\}$ with weakly higher ex ante profit. To see that every optimal distribution must have mass only on these posteriors, observe that the improvement is strict whenever the original distribution assigns positive mass to $(c, 1)$, by Lemma 5, or whenever the original lower tail is not concentrated at zero. Consequently, every seller-optimal information structure must have posterior mass only on posteriors $\{0, c^+, 1\}$.

It remains to prove that the probability mass is optimally positive on each of these posteriors. This is obvious for posteriors c and 1 , but not for 0 . Suppose that $f_0 = 0$. Positive profit and indifference imply

$$f_c = \hat{f}_c = \frac{2 - 3c}{(1 - c)(2 - c)} > 0,$$

which is possible only if $c < 2/3$. Set $f_0 = \varepsilon$ and adjust $f_c(\varepsilon)$, with $f_1(\varepsilon) = 1 - \varepsilon - f_c(\varepsilon)$, to preserve indifference. This adjustment is locally feasible: both remaining masses are positive, and the indifference equation has a nonzero derivative with respect to f_c . Using $\mu_0 = f_1 + cf_c$, differentiation gives

$$f'_c(0) = \frac{(2 - c)(2c - 1)}{(1 - c)(2 - 3c)}.$$

Consequently, at $\varepsilon = 0$,

$$\frac{d}{d\varepsilon} [\mu_0 \pi^1 + (1 - \mu_0) \pi^0] = \frac{c^2(2c - 1)}{(1 - c)(2 - 3c)} > 0,$$

contradicting optimality. Thus $f_0 > 0$ whenever $c > 1/2$. $\square$

Lemma 7

If $c < 1/2$, every seller-optimal information structure F^ is supported on $[c^+, \bar{\mu}]$ for some numerical posterior $\bar{\mu} \in (c, 1]$. It has a positive atom at c^+ and is atomless with full support on $(c^+, \bar{\mu})$.*

Proof. Consider first a problem in which all posteriors trade, i.e., all posterior mass is on $[c^+, 1]$. We verify at the end that this is without loss of optimality when $c < 1/2$.

Let $Q : [0, 1] \rightarrow [c, 1]$ be a quantile function, so that $\mu_0 = \int_0^1 Q(u) du$. Given that all posteriors trade, the ex-ante profit per seller can be written as

$$\begin{aligned} \pi(Q) &= \frac{1}{2} \left((1 - c) \mathbb{E}[\max\{\mu_i, \mu_j\}] - \mathbb{E}[\min\{\mu_i, \mu_j\}] + c \right) \\ &= \frac{1}{2} \left((1 - c) 2 \int_0^1 u Q(u) du - 2 \int_0^1 (1 - u) Q(u) du + c \right) \\ &= \frac{c}{2} + \int_0^1 ((2 - c)u - 1) Q(u) du. \end{aligned} \tag{28}$$

To derive the indifference constraint in terms of the quantile function, consider the profit from choosing high quality. Conditional on high quality, posterior rank u has likelihood $Q(u)/\mu_0$ by Bayes' rule. Given posterior $Q(u)$, the seller makes the sale only when the rival has rank $v < u$, in which case her profit is $Q(u) - Q(v)$. Therefore,

$$\mu_0\pi^1 = \int_0^1 Q(u) \int_0^u [Q(u) - Q(v)] dv du \quad (29)$$

Given (28) and (29), we can express a relaxed version of the indifference constraint as follows:

$$\begin{aligned} \pi^1 \leq \pi^0 &\iff \mu_0(1 - \mu_0)(\pi^1 - \pi^0) \leq 0 \\ \iff \mathcal{I}(Q) &:= \int_0^1 Q(u) \left[uQ(u) - \int_0^u Q(v) dv \right] du - \mu_0\pi(Q) \leq 0 \end{aligned} \quad (30)$$

We can also simplify $\mathcal{I}$ to

$$\mathcal{I}(Q) = \int_0^1 uQ(u)^2 du - \frac{\mu_0^2}{2} - \mu_0\pi(Q).$$

Ignoring any indifference constraint, the unique maximizer of (28) equals c below $u = 1/(2 - c)$ and 1 above it. By Lemma 1, this unconstrained optimizer has $\mathcal{I}(Q) > 0$ when $c < 1/2$. In a constrained problem with the inequality constraint $\mathcal{I}(Q) \leq 0$, the constraint must bind: otherwise we can simply mix towards the unconstrained maximizer, which strictly increases the linear objective and, by continuity, reaches indifference $\mathcal{I}(Q) = 0$. We can hence consider a relaxed problem where the function Q is chosen to maximize (28) subject to the inequality constraint (30) and a monotonicity constraint on Q .

For a moment, consider a relaxed problem where the monotonicity constraint is dropped. The objective (28) is linear, while $\mathcal{I}$ is strictly convex.¹⁸ Therefore, there exists $\lambda > 0$ such that a solution maximizes $\pi(Q) - \lambda\mathcal{I}(Q)$; see (Luenberger, 1969, Sections 8.3–8.4). Setting the marginal payoff $(2 - c)u - 1 - \lambda[2uQ(u) - \pi(Q) - (2 - c)\mu_0u]$ equal to zero when $c < Q(u) < 1$ and setting $Q(u) \in \{c, 1\}$ at the boundaries gives the following first-order condition:

$$Q(u) = \min \left\{ 1, \max \left\{ c, \frac{(2 - c)(1 + \lambda\mu_0)}{2\lambda} - \frac{1 - \lambda\pi(Q)}{2\lambda u} \right\} \right\}. \quad (31)$$

¹⁸To see convexity of $\mathcal{I}$, consider a non-zero perturbation h : we have

$$D^2\mathcal{I}(Q)[h, h] = 2 \int_0^1 uh(u)^2 du + \left(\int_0^1 h(u) du \right)^2 - 2(2 - c) \left(\int_0^1 h(u) du \right) \left(\int_0^1 uh(u) du \right) > 0,$$

where the inequality follows from Cauchy-Schwartz.

Moreover, $1 - \lambda\pi(Q) > 0$; otherwise the right-hand side of (31) would be non-increasing, and hence we would have $\int_0^u Q(v)dv \geq uQ(u)$ for all u , contradicting $\mathcal{I}(Q) = 0$ in (30). Therefore, the solution (31) satisfies monotonicity.

Note also that in the solution (31), $Q(u) = c$ for u sufficiently near 0. Hence, the solution has a positive atom at c , and the solution has exactly the structure claimed by the lemma.

Finally, we can rule out non-trading posteriors. Take any indifferent information structure that has positive mass on $[0, c^-]$. We next move all this mass to posterior c^+ . This leaves the weighted high-quality payoff $\mu_0\pi^1$ unchanged and strictly raises the weighted low-quality payoff $(1 - \mu_0)\pi^0$. It therefore strictly raises ex ante profit and strictly lowers $\pi^1 - \pi^0$. We end up with an information structure supported on $[c^+, 1]$ that has $\pi^1 - \pi^0 < 0$. Hence, for any indifferent information structure that has positive mass on $[0, c^-]$, there is another information structure that yields a strictly higher profit and is feasible in the relaxed optimization problem considered above. Since the optimum of the relaxed problem is feasible for the original full problem, it must yield at least as much profit as this transformed distribution and therefore strictly dominates the original information structure with posterior mass on $[0, c^-]$. $\square$

This concludes the proof of Theorem 1. $\square$

PROOF OF PROPOSITION 2

For a public recommendation $s \in \{A, B\}$, specify

(q_A, q_B)	$(0, 0)$	$(1, 0)$	$(0, 1)$	$(1, 1)$
$\Pr[s = A \mid (q_A, q_B)]$	$\frac{1}{2}$	$\frac{3+c}{4}$	$\frac{1-c}{4}$	$\frac{1}{2}$

where the probability that B is recommended is $1 - \Pr[s = A \mid (q_A, q_B)]$. All probabilities are in $[0, 1]$ for $c \in (0, 1)$. At prior $\mu_0^R = 2c/(1 + c)$, each recommendation has probability $1/2$, and Bayes' rule gives

$$\begin{aligned}\Pr(q_A = 1 \mid A) &= (\mu_0^R)^2 + \frac{3+c}{2}\mu_0^R(1 - \mu_0^R) = \frac{c(3-c)}{1+c} \equiv \hat{\mu}, \\ \Pr(q_B = 1 \mid A) &= (\mu_0^R)^2 + \frac{1-c}{2}\mu_0^R(1 - \mu_0^R) = c.\end{aligned}$$

If B is recommended, the posteriors are reversed. Given posteriors, pricing follows Section 4.2: the recommended seller charges $\hat{\mu}$ and the other seller

charges c . The conditional quality profits are

$$\pi^1 = \frac{\hat{\mu}(\hat{\mu} - c)}{2\mu_0^R}, \quad \pi^0 = \frac{\hat{\mu}(1 - \hat{\mu})}{2(1 - \mu_0^R)}.$$

Since $\hat{\mu} - c = (1 - c)\mu_0^R$ and $1 - \hat{\mu} = (1 - c)(1 - \mu_0^R)$, we get $\pi^1 = \pi^0 = (1 - c)\hat{\mu}/2 = \pi^R > 0$. This furthermore implies that quality is randomized since the indifference condition $\pi^1 = \pi^0$ is fulfilled.

To see why any pure-strategy symmetric equilibrium yields zero profit, note that the willingness to pay is zero at $\mu_0^R = 0$, while homogeneous-product price competition drives profits to zero at $\mu_0^R = 1$. Additionally, full disclosure can never induce positive profits. Under full disclosure a low-quality seller earns zero, whereas a high-quality seller earns $1 - c$ when the rival's quality is low and zero when the rival's quality is high. Hence,

$$\pi^1 = (1 - \mu_0^R)(1 - c) > 0 = \pi^0 \quad \text{for all } \mu_0 \in (0, 1),$$

so that under full disclosure there is no symmetric equilibrium where the sellers would assign positive probability to both qualities. $\square$

PROOF OF PROPOSITION 3

Consider an admissible symmetric equilibrium with positive profit π and write $\mu_0 \in (0, 1)$, which requires sellers to be indifferent between high and low quality, i.e., $\pi^1 = \pi^0$. Conditional on the public signal and the resulting belief (μ_A, μ_B) , let $P(\mu_A, \mu_B)$ denote half the expected sum of sellers' profits, and let $H(\mu_A, \mu_B)$ denote half the expected sum of profits earned by high-quality sellers (not conditional on them having high qualities). Both payoff expressions are symmetric; their values for (μ_A, μ_B) with $\mu_A \geq \mu_B$ are:

$$(P, H) = \begin{cases} (0, 0), & \text{no trade,} \\ ((1 - c)\mu_A/2, \mu_A(\mu_A - c)/2), & \mu_A \geq c \text{ and } \mu_B \leq c, \\ (((1 - c)\mu_A - \mu_B + c)/2, \mu_A(\mu_A - \mu_B)/2), & \mu_A \geq c \text{ and } \mu_B \geq c. \end{cases} \quad (32)$$

The pricing argument in Section 4.2 continues to apply after conditioning on posterior realization (μ_A, μ_B) . For any interior posterior, Bayes' rule and our refinements require both quality types to use the same price distribution. Hence, although qualities may be correlated conditional on the signal, prices convey no additional information about either product on the equilibrium path.

At posterior c , we allow either trading continuation. Randomization between these continuations preserves the inequalities below because they are linear in payoffs.

Symmetry, Bayes plausibility and quality indifference give

$$\mathbb{E} P = \pi, \quad \mathbb{E} H = \mu_0 \pi, \quad \mathbb{E}[(\mu_A + \mu_B)/2] = \mu_0. \quad (33)$$

We now derive an upper bound for equilibrium profit for each prior μ_0 . Writing $z = c + (1 - c)\mu_0$, we will show that

$$\pi \leq \min \left\{ \frac{(1 - c)z}{2}, \frac{(1 - \mu_0)z}{2 - \mu_0} \right\} \quad (34)$$

First bound. First, we will show

$$\pi \leq \frac{(1 - c)z}{2}. \quad (35)$$

Denote $\lambda = (1 - c)/z$. We show a stronger pointwise version of (35):

$$(1 + \lambda\mu_0)P - \lambda H \leq (1 - c)z/2. \quad (36)$$

When trade occurs with $\mu_A \geq c \geq \mu_B$, the left side of (36) equals $\lambda\{z^2 - (\mu_A - z)^2\}/2$ which is at most $\lambda z^2/2 = (1 - c)z/2$. When trade occurs with $c \leq \mu_B \leq \mu_A$, it is affine in μ_B , so it suffices to check the two endpoints. The endpoint $\mu_B = c$ is covered above by the case $\mu_B \leq c$. At the other endpoint $\mu_A = \mu_B$, it equals $(1 + \lambda\mu_0)c(1 - \mu_A)/2$, which is maximized over $\mu_A \geq c$ at $\mu_A = c$, which again falls under the case $\mu_B = c$ and therefore satisfies the bound. With no trade it is zero. Therefore, we obtain (36), and averaging and using (33) yields (35).

Second bound. Second, we will show

$$\pi \leq \frac{(1 - \mu_0)z}{2 - \mu_0}. \quad (37)$$

Set $\lambda = (2 - c)/\{z(2 - \mu_0)\}$. We claim a stronger pointwise version of (37):

$$(1 + \lambda\mu_0)P - \lambda H \leq \frac{z}{2 - \mu_0} \left(1 - \frac{\mu_A + \mu_B}{2} \right). \quad (38)$$

When trade occurs with $c \leq \mu_B \leq \mu_A$, the right side minus the left is affine in μ_B . Its endpoint values are

$$\frac{\lambda}{2}(\mu_A - z)^2 \quad (\mu_B = c), \quad \frac{\lambda(z - c)^2(1 - \mu_A)}{2(1 - c)} \quad (\mu_A = \mu_B),$$

both nonnegative. Lowering μ_B below c , while holding $\mu_A \geq c$ fixed, leaves the left side unchanged and increases the right side. With no trade the inequality is immediate. Hence we obtain (38), and averaging and using (33) gives (37).

Maximization and equality. We have thus obtained an upper bound (34) for the equilibrium profit as a function of the prior. The two expressions in (34) coincide at $\mu_0^R = 2c/(1+c)$. The first is strictly increasing and is the smaller expression below μ_0^R . For $\mu_0 \geq \mu_0^R$ and $c \geq 1/2$, we have $\mu_0 \geq 2/3$, so

$$\frac{d}{d\mu_0} \left[\frac{(1-\mu_0)z}{2-\mu_0} \right] = (1-c) - \frac{2-c}{(2-\mu_0)^2} \leq \frac{1}{2} - \frac{9}{16} < 0.$$

Hence the upper bound is uniquely maximized at μ_0^R , where it equals

$$\frac{c(1-c)(3-c)}{2(1+c)}.$$

The ranking experiment attains this profit, as established in the proof of Proposition 2, and is therefore seller-optimal. $\square$

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